FOOD DELIVERY TIME PREDICTION

Machine Learning Project Report

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Course : **Machine Learning**

1. Introduction

Food delivery services rely on efficient logistics and accurate time estimation to maintain customer satisfaction. Predicting delivery time is a complex problem influenced by multiple factors such as traffic conditions, distance, weather, and delivery personnel experience.

This project aims to analyse factors affecting delivery time and build predictive models using machine learning techniques.

Two modelling approaches were implemented:

- **Linear Regression** to predict delivery time (continuous variable)
- **Logistic Regression** to classify deliveries as **Fast or Delayed**

The project includes data preprocessing, feature engineering, exploratory data analysis, model training, evaluation, and interpretation of results.

2. Dataset Description

The dataset used in this project is **Food_Delivery_Time_Prediction.csv**, which contains information related to food delivery operations.

Total records: **200**

Important features include:

Feature	***Description***
Customer_Location	Latitude and longitude of customer
Restaurant_Location	Latitude and longitude of restaurant
Distance	Delivery distance
Weather_Conditions	Weather during delivery
Traffic_Conditions	Traffic level
Vehicle_Type	Delivery vehicle
Delivery_Person_Experience	Years of experience
Restaurant_Rating	Rating of restaurant

Customer_Rating	Customer rating
Order_Cost	Cost of order
Tip_Amount	Tip amount
Delivery_Time	Target variable

3. Data Preprocessing

3.1 Data Inspection

Initial inspection was performed to understand dataset structure, data types, and missing values.

Code operations included:

- Checking dataset shape
- Inspecting column data types
- Verifying missing values

Result:
No missing values were detected in the dataset.

3.2 Latitude and Longitude Extraction

The dataset contained location coordinates as strings. These were separated into numeric latitude and longitude features.

New features created:

- cust_lat
- cust_lon
- rest_lat
- rest_lon

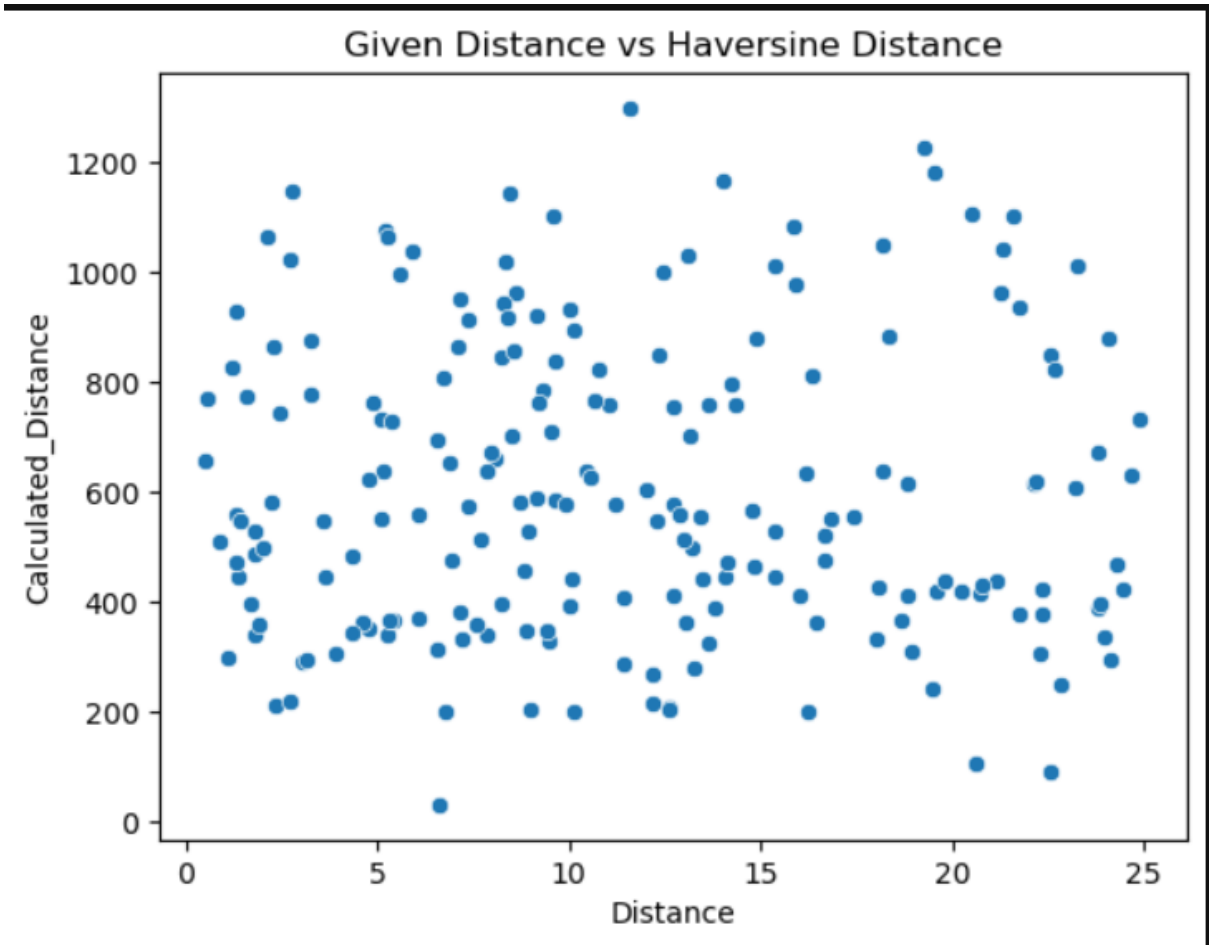
3.3 Distance Validation Using Haversine Formula

To validate the accuracy of the provided distance values, the **Haversine formula** was used to compute geographical distance between customer and restaurant locations.

This formula calculates the shortest distance between two points on Earth based on latitude and longitude.

A new feature **Calculated_Distance** was generated.

Plot: Given Distance vs Calculated Distance



Observation:

The calculated distances were significantly larger than the distances provided in the dataset. This suggests that the latitude and longitude values in the dataset may not represent realistic delivery locations.

Therefore:

- The original **Distance feature was retained for modelling.**
- The calculated distance was used only for validation purposes.

4. Feature Engineering

To improve model performance and capture hidden relationships, several new features were created.

4.1 Rush Hour Feature

Delivery demand typically increases during morning and evening periods.

A binary feature **Rush_Hour** was created:

- Morning / Evening → Rush Hour (1)
- Afternoon / Night → Non-Rush Hour (0)

This feature captures peak traffic periods.

4.2 Traffic–Distance Interaction

Traffic congestion impacts longer routes more severely than shorter ones.

An interaction feature was created:

$$\text{Traffic_Distance} = \text{Traffic_Level} \times \text{Distance}$$

This captures the combined effect of traffic and distance on delivery time.

4.3 Experience Efficiency

Delivery personnel experience can affect travel efficiency.

Feature created:

$$\text{Experience_per_KM}$$

This measures experience relative to travel distance.

4.4 Cost per Kilo meter

Order cost may influence delivery priority or service quality.

Feature created:

$$\text{Cost_per_KM} = \text{Order_Cost} / \text{Distance}$$

5. Categorical Encoding

Categorical variables were converted into numerical form using **One-Hot Encoding**.

Encoded features include:

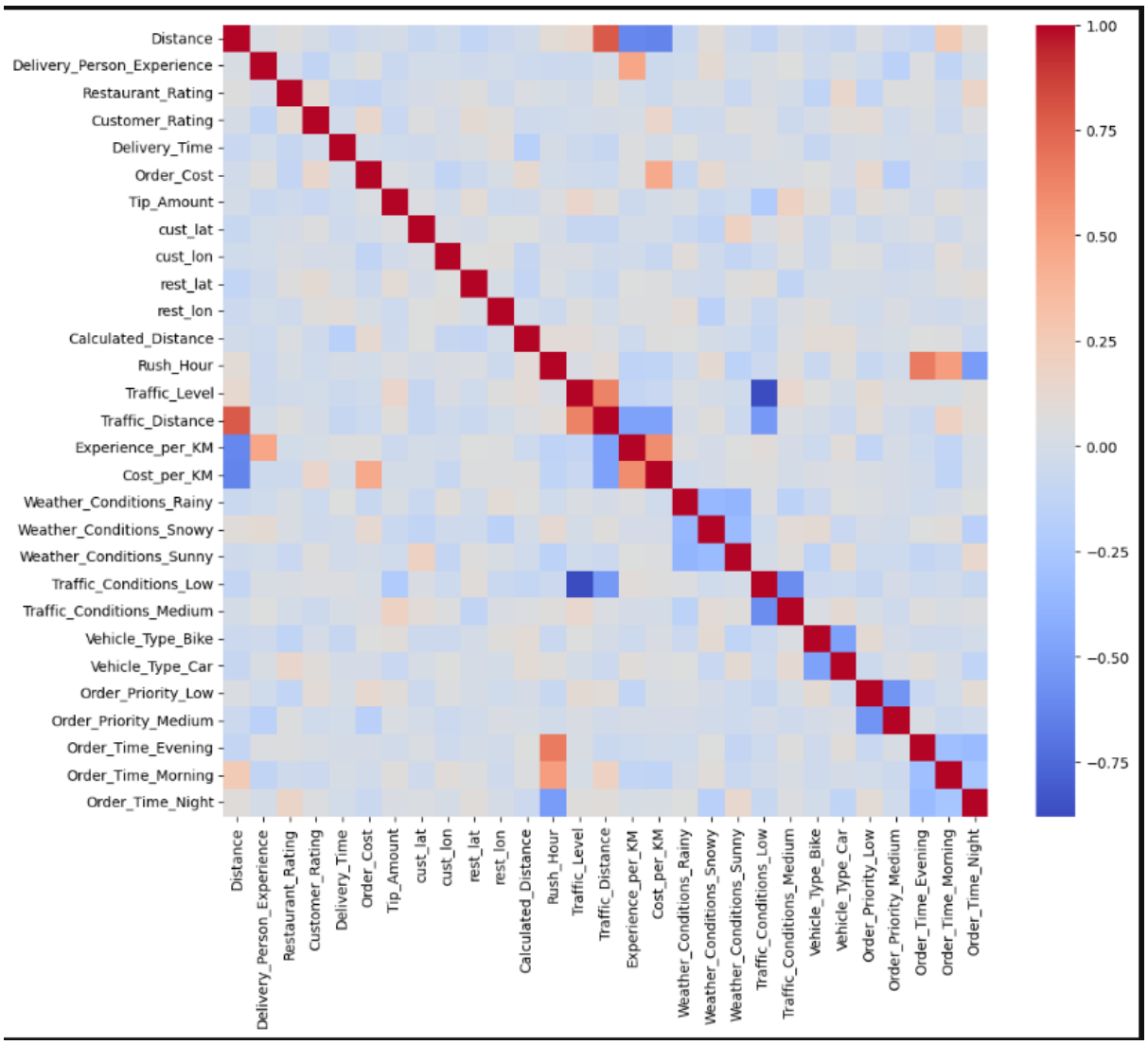
- Weather Conditions

- Traffic Conditions
- Vehicle Type
- Order Priority
- Order Time

To prevent multicollinearity, the first category was dropped.

6.1 Correlation Analysis

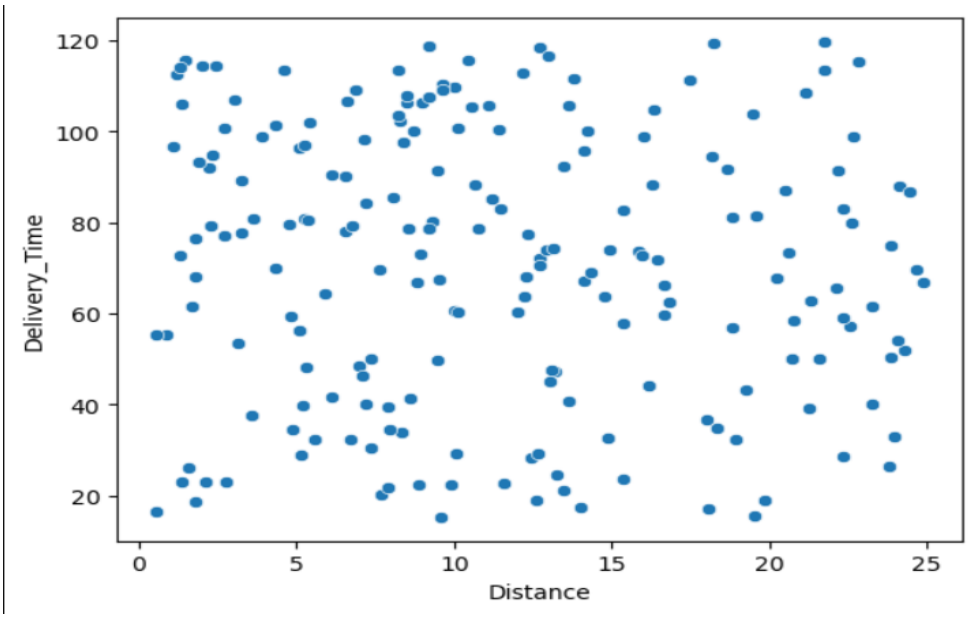
A correlation heatmap was generated to identify relationships between variables.



Observation:

Most variables show weak correlation with delivery time, indicating that delivery time may depend on complex interactions between features rather than single variables.

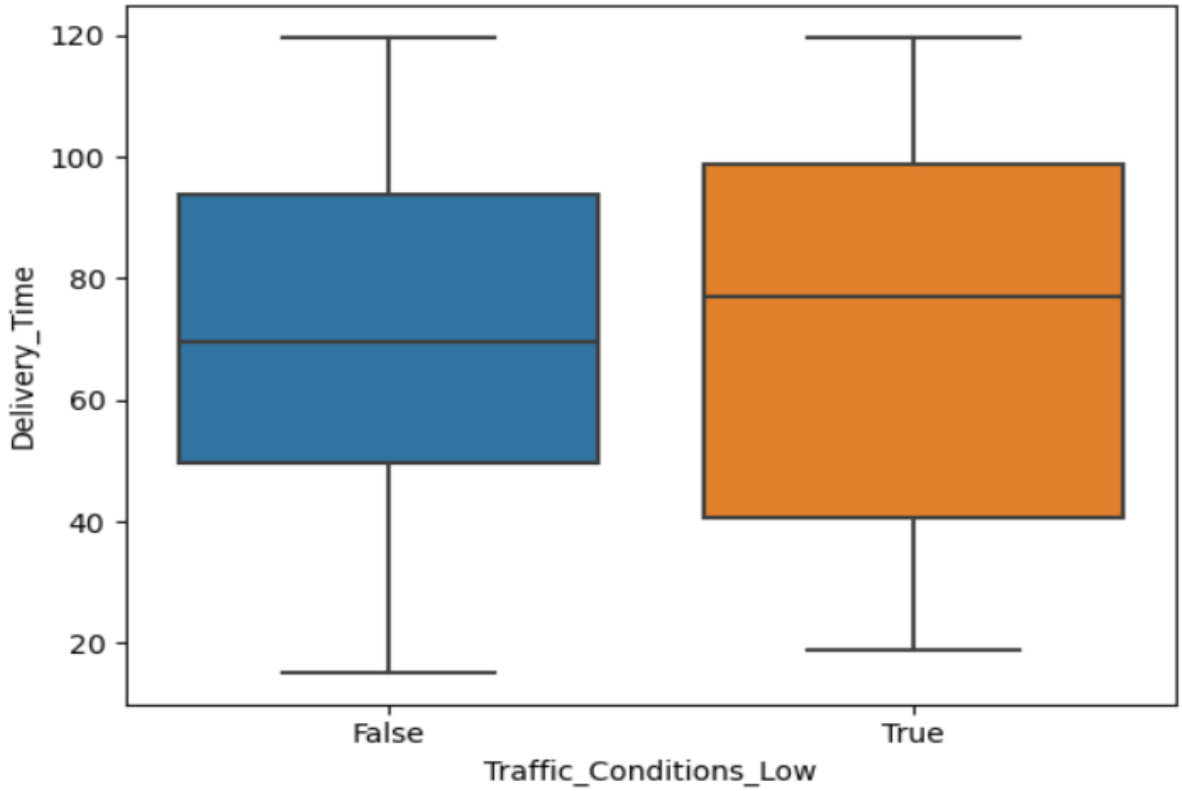
6.2 Distance vs Delivery Time



Observation:

No strong linear trend was observed between distance and delivery time.

6.3 Traffic Impact

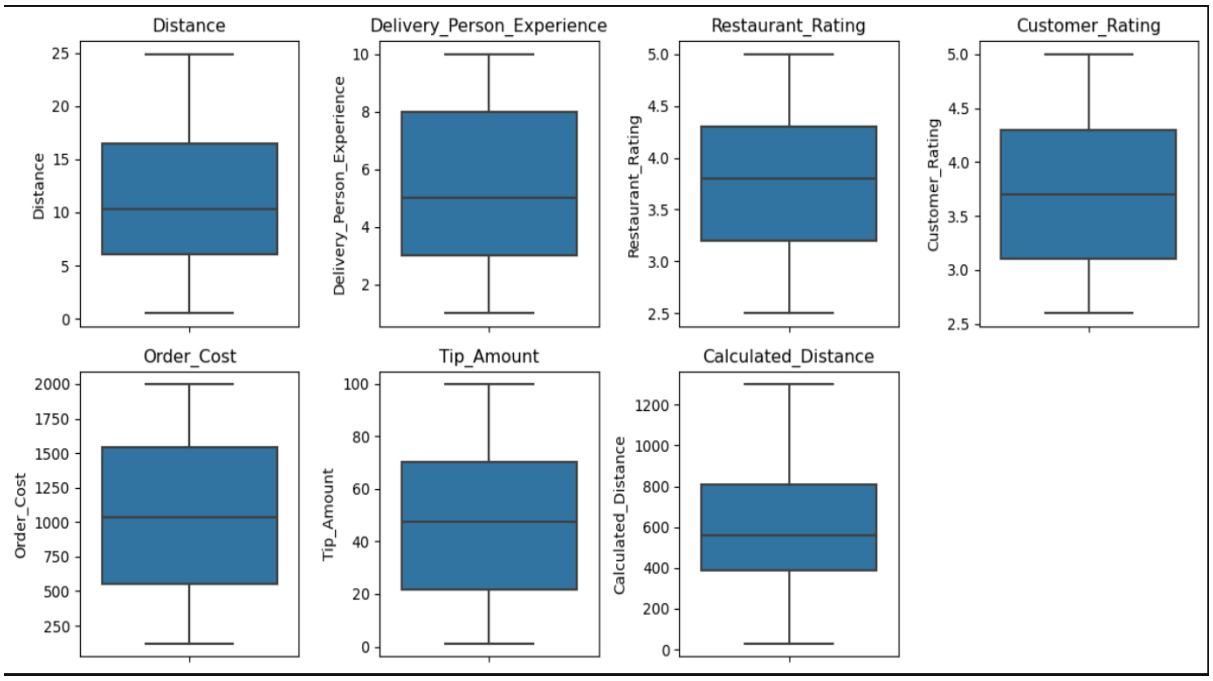


Observation:

Traffic conditions slightly affect delivery time, but significant overlap exists between categories.

7. Outlier Detection

Outliers were analysed using boxplots.



Extreme values were evaluated to ensure they did not distort model predictions.

8. Train-Test Split and Feature Scaling

The dataset was divided into:

- **80% training data**
- **20% testing data**

Numeric features were standardized using Standard Scaler to ensure consistent scale across predictors.

9. Linear Regression Model

A Linear Regression model was trained to predict delivery time.

Evaluation metrics used:

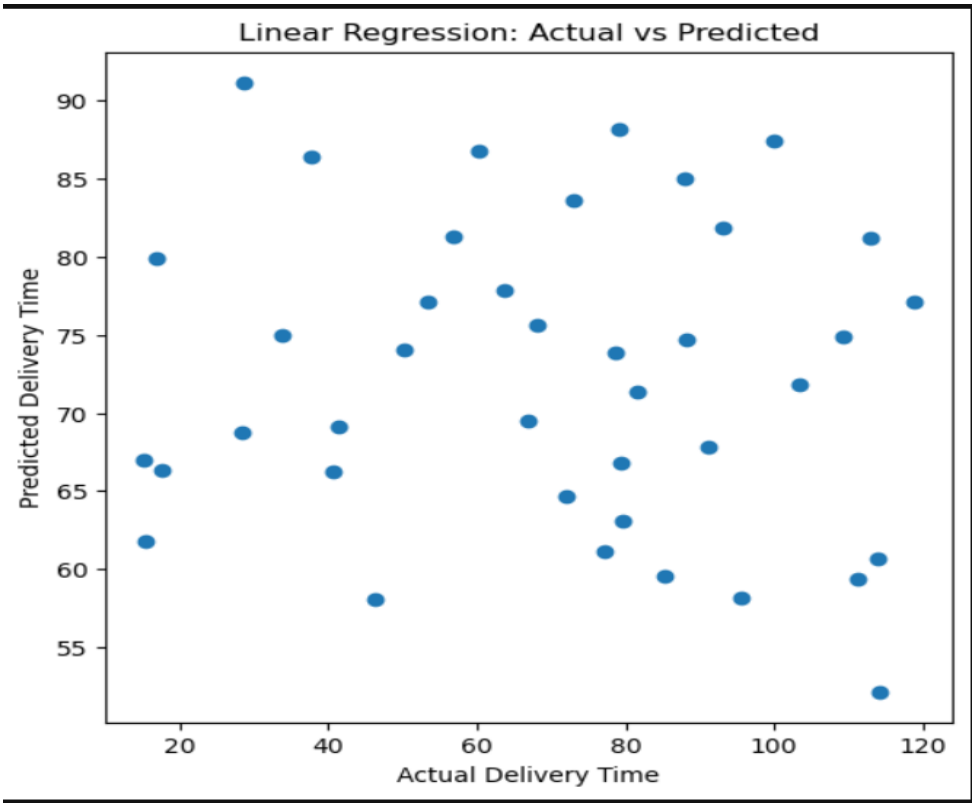
- **Mean Squared Error (MSE)**
- **Mean Absolute Error (MAE)**
- **R^2 Score**

Linear Regression Results

```
Linear Regression Results
MSE: 1087.12742233244
MAE: 27.7799316702047
R²: -0.175411130778947
```

Observation:

The model produced a negative R^2 score, indicating poor predictive performance.



10. Model Improvement Attempts

To address the poor performance of Linear Regression, several improvement strategies were attempted.

10.1 Feature Engineering

New interaction features such as Rush_Hour and Traffic_Distance were introduced.

10.2 Feature Selection

Correlation analysis was used to identify weak predictors.

Variables with extremely low correlation were considered less influential.

10.3 Multicollinearity Analysis

Variance Inflation Factor (VIF) was calculated to detect multicollinearity between predictors.

High VIF values indicate redundancy among variables.

10.4 Polynomial Regression

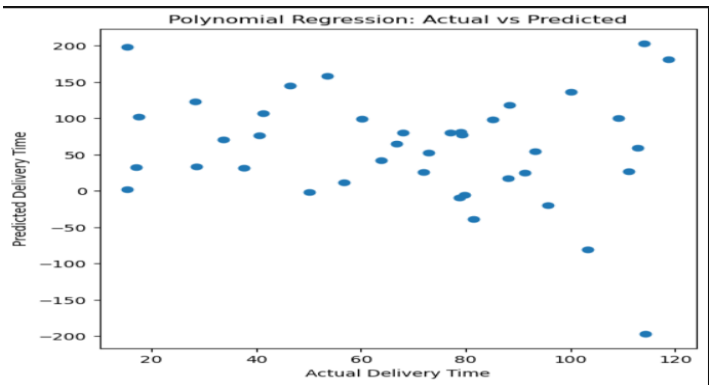
Polynomial features (degree = 2) were introduced to capture nonlinear relationships between variables.

Polynomial Regression Results

```
Polynomial Regression Results
MSE: 7390.236587280999
MAE: 61.165794138216576
R²: -6.9903847197072935
```

Observation:

Polynomial regression slightly improved model flexibility but overall performance remained limited due to weak feature relationships.



11. Logistic Regression Model

To perform classification, delivery times were converted into binary categories:

- **Fast Delivery**
- **Delayed Delivery**

The threshold was determined using the median delivery time.

12. Logistic Regression Evaluation

Metrics used:

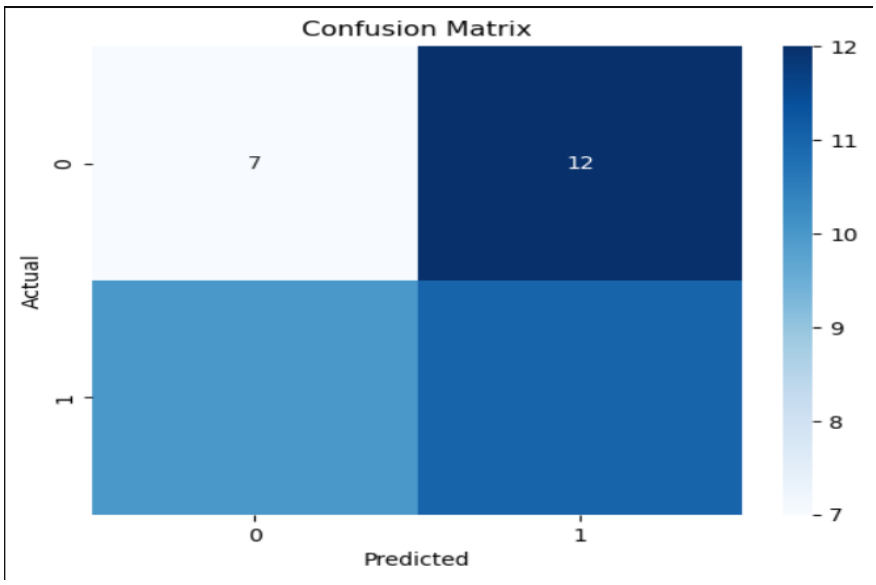
- **Accuracy**
- **Precision**
- **Recall**
- **F1 Score**

Logistic Regression Results

```
Accuracy: 0.45
Precision: 0.4782608695652174
Recall: 0.5238095238095238
F1 Score: 0.5
```

These results indicate that the model performs close to random guessing.

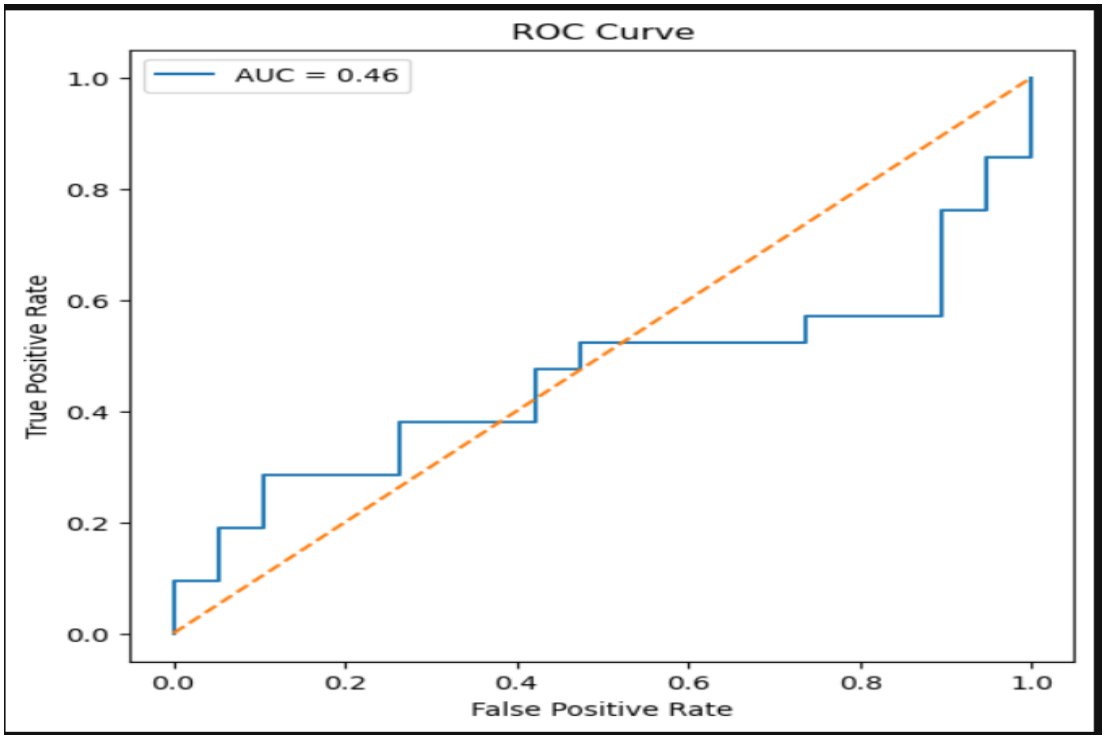
13. Confusion Matrix



Interpretation:

The confusion matrix shows the distribution of correct and incorrect classifications.

14. ROC Curve



The ROC curve evaluates the model's ability to distinguish between fast and delayed deliveries.

The model shows limited classification capability.

15. Model Comparison

Model	Task	Performance
Linear Regression	Predict delivery time	Negative R^2
Polynomial Regression	Capture nonlinear patterns	Slight improvement
Logistic Regression	Classify fast/delayed	~50% accuracy

Overall, the models demonstrate weak predictive power due to limited feature relationships.

16. Key Findings

- **Delivery time shows weak correlation with individual features.**
- **Distance alone is not a strong predictor.**
- **Traffic and weather influence delivery time but are insufficient for accurate prediction.**
- **The dataset may lack important operational variables.**

18. Conclusion

This project demonstrated the full machine learning pipeline for delivery time prediction. Although model performance was limited, the analysis highlighted the importance of high-quality features and operational data in predictive modelling.

Further improvements could be achieved by incorporating real-world logistics data and advanced machine learning models such as Random Forest or Gradient Boosting.